

FreshBasket

BCA Final Year
Project
2025-26

Presented by
BCA Final Year
Students

Institution
Online Fruits and Vegetable Delivery
Platform

Fresh fruits and vegetables, delivered to your door

FreshBasket is a full-stack platform for ordering fruits, vegetables, salad-bowls , juices with secure checkout and real-time order tracking.





Traditional grocery shopping is overdue for a fix

01

Time consuming

Commuting and browsing long aisles eats up hours every week.

02

Long billing queues

Peak-hour checkout lines delay even a simple grocery run.

03

Limited store hours

Fixed hours block access to essentials after work or on weekends.

04

No price visibility

Comparing prices across stores requires visiting each one in person.

What FreshBasket delivers

Secure auth

User registration and JWT-based login with session management

Smart search

Category browsing, keyword search, and advanced product filters

Cart management

Add, remove, update quantities, and apply coupon codes

Order tracking

End-to-end order placement, live status updates, and invoice generation

Payment gateway

Secure online payments via Razorpay integration

Admin dashboard

Product, order, and user management with analytics

Technology stack

Built on a modern, scalable stack

A proven combination of technologies ensures performance, security, and maintainability. Coverage spans the full product - from UI to database.

Frontend

React.js, HTML5, CSS3, and JavaScript for a responsive, dynamic user interface.

Backend

Node.js and Express.js power a clean, scalable REST API layer.

Database

MySQL with a structured relational schema for reliable data management.

Auth & Tools

JWT auth, Razorpay payments. Built with Git, Postman, and VS Code.

Five core modules

The building blocks of FreshBasket

User

Registration, login, profile management, and order history

Products

Category browsing, search, filters, and product detail pages

Cart

Add or remove items, update quantities, and apply discount coupons

Orders

Placement, real-time tracking, and invoice generation

Admin

Product and order management, analytics dashboards, stock management and waste tracking

Notifications

Real-time alerts for order updates

1 User interface

React.js SPA delivers a responsive shopping experience across all devices

2 Backend API

Node.js and Express.js process requests, apply business logic, and return responses

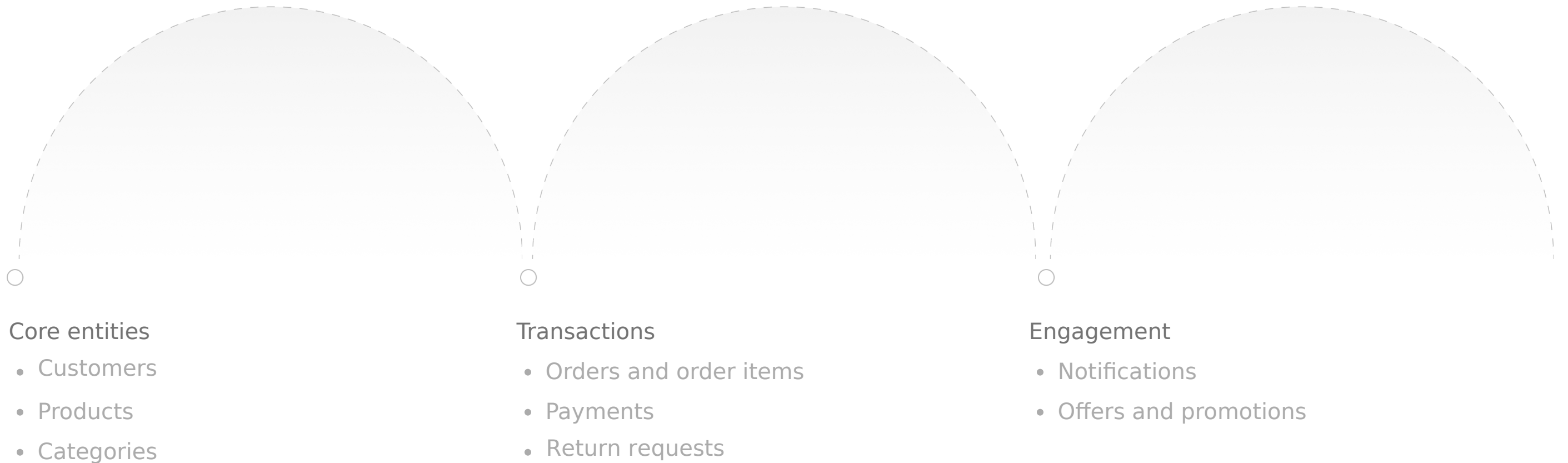
3 MySQL database

Stores user accounts, products, orders, payments, and transaction records securely

4 Payment gateway

Razorpay handles secure payment processing and returns confirmation to the backend

Relational schema built for scale



Built for users

Six features that put shoppers first

Secure login

JWT sessions and bcrypt hashing
keep every account protected

Returns

Raise return or replacement
requests from order history

Reward points

Earn points on every order,
redeemable for future discounts

Coupons

Apply discount codes at checkout
for instant savings

Notifications

Real-time alerts for order status,
delivery, and offers

Powerful admin control center

Full visibility and control over products, orders, and platform performance from a single dashboard.

Insights

Analytics

Sales dashboards and visual reports showing revenue, top products, and trends.

Products

Inventory

Real-time stock management with low-stock alerts and full product CRUD.

Users

User management

View and manage customer accounts, orders, returns, and platform settings.

Secured and tested end to end

Multi-layered security and thorough testing ensure a reliable, trustworthy platform for every user.

Stateless tokens secure all API endpoints and protect user sessions

Passwords are hashed before storage and never exposed in plain text

All REST endpoints validated with Postman for correctness and edge cases

Unit and integration tests verify core business logic and module interactions

Future scope

Where FreshBasket goes next

AI recommendations

Smart product suggestions based on individual purchase history and preferences.

Mobile app

Native iOS and Android apps for seamless on-the-go grocery ordering.

Multi-language

Platform localization for regional Indian languages to reach wider audiences.

Delivery partners

Integration with third-party last-mile delivery logistics APIs for faster fulfilment.

FreshBasket



- Full-stack platform: React.js frontend, Node.js/Express.js backend, and MySQL database
- Secure architecture: JWT authentication, bcrypt hashing, and Razorpay payment integration
- Complete modules: User, product, cart, order, and admin management in a single system
- Scalable schema: Relational database design built to grow with users and transactions

Thank You